Hockey, AI, & the future of sports analysis!

Podcast

Summary

In this episode of Approximately Correct, hosts Alona Fyshe and Scott Lilwall talk with David Radke, a senior research scientist with the Chicago Blackhawks and a PhD graduate of the University of Waterloo, about how modern hockey data is colliding with reinforcement learning ideas. During graduate school, he used NHL tracking data to develop AI-driven analytics that measured player performance in new ways, including passing behavior and on-ice positioning.

Radke explains why "invasion games" like hockey and soccer are harder to model than sports like baseball. In invasion games, the game state is constantly changing because every player's position matters, even when they do not have the puck. He describes two main kinds of hockey data: event data (logged actions like passes with coordinates and time, but without full context of where everyone is) and newer tracking data, which records the locations of all players and the puck multiple times per second using sensors. He notes this tracking system was implemented league-wide after first being used during the 2020 "bubble," and that all teams receive access to the same dataset through the league.

From there, the conversation connects hockey analytics to multi-agent reinforcement learning. Radke describes how teams of agents can learn to coordinate from reward signals, and why reward design becomes tricky when many agents share the same outcome. A central idea is the credit assignment problem: if the reward is too global and too delayed, agents may not get a strong enough signal to learn which of their actions mattered. In his research, he found that fully cooperative "everyone shares the same reward" setups can underperform compared to systems that use smaller subgroups or mixed incentives (a blend of self-interest and group outcomes), because these designs create clearer feedback signals.

In fully cooperative systems, every agent shares the same reward, which can make it difficult for individuals to see how their own actions affect the outcome. This can lead to what researchers call "lazy agents," not because the agents are unmotivated, but because the feedback signal is too weak for them to learn what actually helped. By breaking a large population into smaller subgroups, Radke found that agents receive clearer feedback, improve credit assignment, and that overall performance can surpass fully cooperative systems.

Finally, Radke describes how these ideas show up in sports analytics today. Since goals are rare, hockey is a sparse-reward environment. Instead of only valuing the goal itself, analysts try to estimate the value of actions that lead up to scoring, such as the value of a pass in a given situation. He discusses approaches that resemble inverse reinforcement learning, where you learn a value function from historical behavior (offline game data) by working backward from outcomes. He also points to a big open challenge: in invasion sports, individual performance can depend heavily on linemates and team systems, which complicates single-number player metrics like wins above replacement (WAR).

Key Terms

Term	Definition
Agent	An agent in reinforcement learning (RL) is an entity that learns to make decisions by interacting with an environment.
Credit assignment problem	The challenge of figuring out which actions actually caused a reward, especially when rewards are delayed or shared across many players or agents.
Event data	A record of every sport action like passes and shots with timestamps and context. This fundamental data provides deep insights into player performance, team strategies, and game dynamics.
Invasion game	A sport where teams attack into the opponent's territory to score (for example, hockey or soccer), creating constant multi-player interaction and a complex game state.
Intransitive strategy space	A situation where strategy matchups can form cycles (Team A beats Team B, Team B beats Team C, Team C beats Team A), so "best strategy" depends on the opponent.
Inverse reinforcement learning	Where an agent learns the value of states or actions by observing existing behavior and outcomes, rather than learning through trial and error as in traditional reinforcement learning.

Multi-agent reinforcement learning	Reinforcement learning where multiple agents learn and interact in the same environment, often requiring coordination or competition.
Reinforcement learning	A type of machine learning where an agent learns optimal behavior by interacting with an environment through trial and error.
Reward signal	The feedback used to guide learning (or evaluation), which can be simple in principle but hard to design well in complex, multi-agent settings.
Tracking data	Sensor-based or vision-based data that records player and puck positions frequently over time, providing a fuller picture of the "state" of the game.
Value function	An estimate of how good a situation is, or how beneficial an action is, in terms of expected future outcomes (for example, how a pass increases the chance of scoring later).

3 Key Takeaways

1. Hockey is a decision-heavy, multi-agent problem.
2. Better feedback signals can beat "pure cooperation."
3. Tracking data changes what is measurable.

Discussion Questions

1. Think about a project you have worked on where success was measured only at the end (a final deliverable, quarterly target, launch day). What "missing signals" made it hard to know what was working along the way, and what smaller or more immediate metrics could have helped?

2. Radke suggests teams are "more than the sum of their parts" because complementary roles matter. Where in your work do complementary skills create better outcomes than having everyone do the same "high value" task?

3. In hockey, goals are rare, so analysts look for ways to value actions that lead up to goals. In your context, what are the "goals," and what upstream actions might deserve more credit even if they do not look impressive on their own?

Relevant Resources

For more information on topics mentioned in this podcast, check out:

- For learners interested in the theory behind the episode's discussion of cooperation and incentives:
  - Cooperative AI: Machines Must Learn to Find Common Ground by Kate Larson
- On reinforcement learning:
  - Introduction to Reinforcement Learning with MatLab
  - Introduction to Multiagent Reinforcement Learning
- On sports analytics:
  - Coursera's How is AI Being Used in Sports Analytics Today
  - Google DeepMind's TacticAI: an AI assistant for football tactics

References

Alberta Machine Intelligence Institute. (2025, March 14). Hockey, AI, & the future of sports analysis!| Approximately Correct podcast [Video]. YouTube. https://www.youtube.com/watch?v=hiWQiTynYyU

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